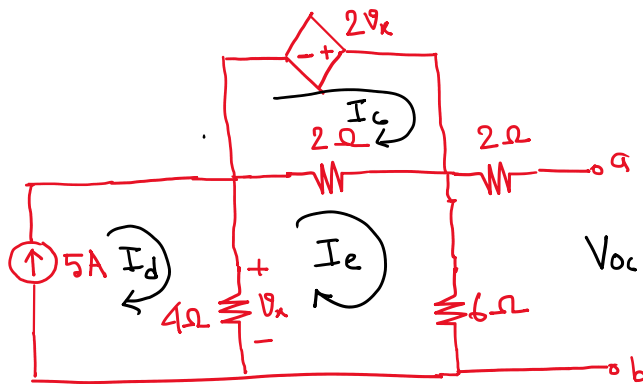
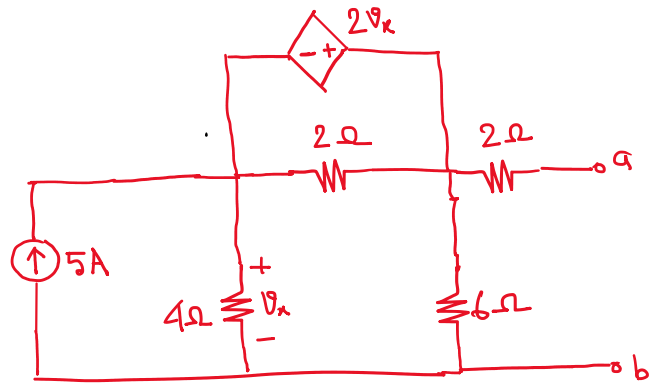
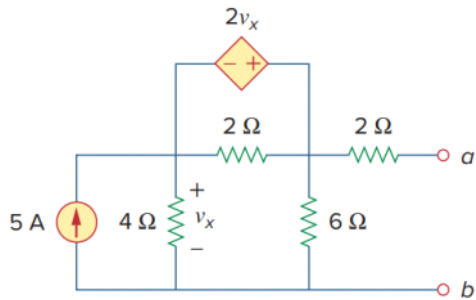


Thevenins_Dependent source

Monday, December 9, 2024 8:10 PM



$$V_x = 4(I_d - I_e)$$

Apply mesh Analysis:

In loop d:

$$I_d = 5A$$

In loop e:

$$-V_4 - V_2 - V_6 = 0$$

$$\Rightarrow -4(I_e - I_d) - 2(I_e - I_c) - 6I_e = 0$$

$$\Rightarrow +4I_d - 12I_e + 2I_c = 0$$

$$\Rightarrow 4 \times 5 - 12I_e + 2I_c = 0$$

$$\Rightarrow 12I_e - 2I_c = 20 \quad \text{--- (i)}$$

In loop c:

$$+2V_x - 2(I_c - I_e) = 0$$

$$\Rightarrow 2V_x - 2I_c + 2I_e = 0$$

$$\Rightarrow 2 \times 4(I_d - I_e) - 2I_c + 2I_e = 0$$

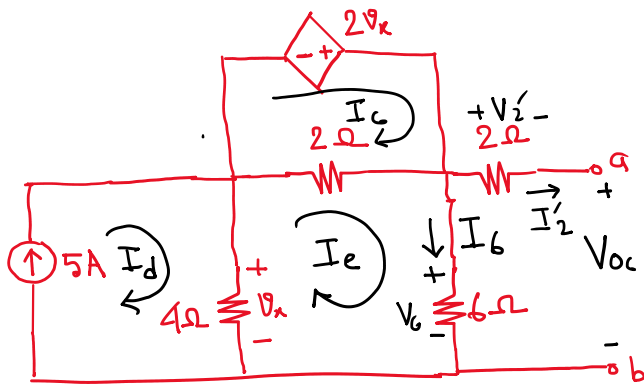
$$\Rightarrow 8I_d - 8I_e - 2I_c + 2I_e = 0$$

$$\Rightarrow 8 \times 5 - 2I_c - 6I_e = 0$$

$$\Rightarrow 2I_c + 6I_e = 40 \quad \text{--- (ii)}$$

Solving the equations we find:

$$I_e = 10/3 A$$

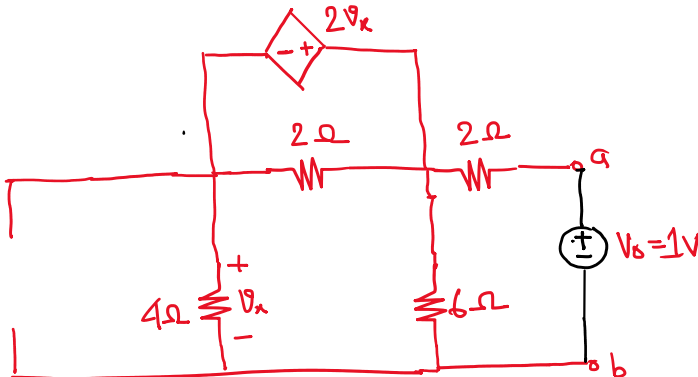


Applying KVL:

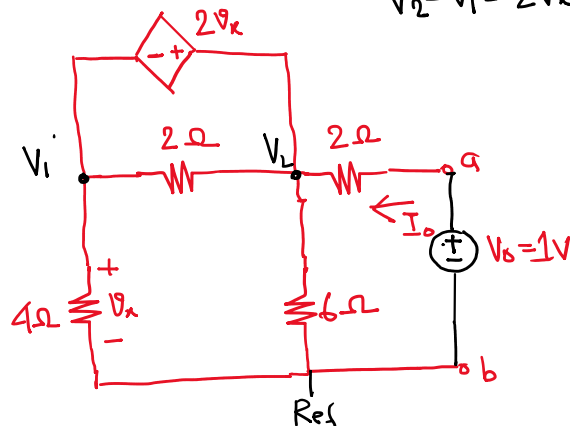
$$\begin{aligned}
 +V_6 - V_2' - V_{oc} &= 0 \\
 \Rightarrow +I_c \times 6 - 2I_2' - V_{oc} &= 0 \\
 \Rightarrow 6 \times \frac{10}{3} - 0 - V_{oc} &= 0 \quad [I_6 = I_e = 10/3 \text{ A} \text{ \& } I_2' = 0] \\
 \Rightarrow V_{oc} &= 20 \text{ V}
 \end{aligned}$$

So, $V_{TH} = V_{oc} = 20 \text{ V}$

- To determine R_{TH} :
- 1 Remove the independent source
 - 2 Keep the dependent sources.
 - 3 Connect a voltage source $V_0 = 1 \text{ V}$ betⁿ a & b



$$V_2 - V_1 = 2V_x$$



Applying node Analysis:

$$\begin{aligned}
 \frac{V_1}{4} + \frac{V_1 - V_2}{2} + \frac{V_2 - V_1}{2} + \frac{V_2 - 0}{6} + \frac{V_2 - 1}{2} &= 0 \\
 \Rightarrow \frac{V_1}{4} + \frac{V_2 + 3V_2 - 3}{6} &= 0 \\
 \Rightarrow \frac{3V_1 + 8V_2 - 6}{12} &= 0
 \end{aligned}$$



$$@ \quad V_L = V_1$$

$$\Rightarrow 2V_L = 2V_1 = V_2 - V_1$$

$$\Rightarrow 3V_1 = V_2$$

$$\Rightarrow \frac{3V_1 + 8V_2 - 6}{12} = 0$$

$$\Rightarrow 3V_1 + 8V_2 - 6 = 0$$

$$\Rightarrow 3V_1 + 8V_2 = 6$$

$$\Rightarrow V_2 + 8V_2 = 6$$

$$\Rightarrow 9V_2 = 6 \Rightarrow V_2 = \frac{6}{9} = \frac{2}{3}$$

$$I_o = \frac{V_o - V_2}{2} = \frac{1 - \frac{2}{3}}{2} = \frac{\frac{1}{3}}{2} = \frac{1}{6} \text{ A}$$

$$R_{TH} = \frac{V_o}{I_o} = \frac{1}{1/6} = 6 \Omega$$